

Project Report: Car Price Prediction with Machine Learning

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Tools: Python, Pandas, NumPy, Scikit-Learn, Matplotlib/Seaborn

1. Project Overview

The objective of this project is to develop a predictive model that estimates the selling price of used cars. Car price prediction is a complex task because the resale value depends on a multitude of factors, including brand reputation, age, fuel type, and mileage. This project leverages the **Random Forest Regressor algorithm** to provide accurate, data-driven pricing insights.

2. Dataset Analysis

The project utilizes the "**car data.csv**" dataset, which consists of 301 records and 9 features.

Key Features:

- **Car_Name:** The model/brand of the car.
- **Year:** The year of manufacture.
- **Present_Price:** The current showroom price of the car (in Lakhs).
- **Selling_Price (Target):** The price at which the car is being sold.
- **Driven_kms:** Total distance traveled by the car.
- **Fuel_Type:** Petrol, Diesel, or CNG.
- **Selling_type:** Whether sold by a Dealer or an Individual.
- **Transmission:** Manual or Automatic.
- **Owner:** Number of previous owners.

3. Data Preprocessing & Feature Engineering

To prepare the data for machine learning, several steps were taken:

- **Handling Missing Values:** The dataset was verified to have zero missing values.
- **Feature Engineering:** A new feature, Car_Age, was created by subtracting the manufacturing Year from the current year (2026).
- **Categorical Encoding:** Non-numeric features (Fuel_Type, Selling_type, and Transmission) were converted into numeric format using One-Hot Encoding.
- **Feature Selection:** The Car_Name and original Year columns were dropped to prevent noise and overfitting.

4. Exploratory Data Analysis (EDA)

Visualizations were used to understand feature relationships:

- **Scatter Plot:** Revealed a strong positive linear correlation between Present_Price and Selling_Price.
- **Correlation Heatmap:** Confirmed that Present_Price is the most significant predictor (0.88 correlation), while Car_Age and Driven_kms show a negative correlation with price.

5. Model Development

The data was split into a training set (80%) and a testing set (20%).

- **Algorithm:** Random Forest Regressor.
- **Reasoning:** Random Forest is an ensemble method that handles non-linear relationships effectively and reduces the risk of overfitting compared to a single decision tree.

6. Performance Evaluation

The model's performance was measured using standard regression metrics:

- **R² Score: 0.96** (The model explains 96% of the variance in car prices).

- **Mean Absolute Error (MAE): 0.64 Lakhs** (On average, predictions are within 0.64 units of the actual price).

Feature Importance Ranking:

- 1. Present Price (88.1% influence)**
- 2. Car Age (5.9% influence)**
- 3. Driven Kms (4.0% influence)**

7. Conclusion

The Car Price Prediction model is highly successful, with an accuracy rate of 96%. The analysis proves that the initial cost of the car and its age are the primary drivers of depreciation. This tool can be effectively used by dealerships or individuals to estimate fair market values for used vehicles.